PHAS0040

Nuclear and Particle Physics

Thorough Revision Guide

Built from analysis of all 5 past papers (2020/21 → 2024/25),

Assessed Problem Sets 1–3, PSTs 1–5, and the 2025/26 write-up.

1. How this guide was built (read this first)

Your write-up is ~145 pages and reads like a textbook, not lecture notes. The reason it feels overwhelming is simple: most of it will not be assessed. The course is mostly memorisation-heavy, with a few specific calculation patterns that get recycled almost every year. The job of revision is to figure out which 30% matters and ignore the rest.

To build this guide I went through every past paper question from 2020/21 to 2024/25, every assessed problem sheet (Problem Set 1, 2, 3) and every PST (1–5), and tagged each question against the 10 chapters of the write-up. Topics that appear in 4 or 5 papers are 'must know'. Topics that appear in 1 paper are 'skim'. The priority matrix in Section 3 is the result, and inside every Section 4 chapter you'll find a blue box listing the exact questions that test that chapter.

This guide is designed to be used with your existing materials, not to replace them. The sequence is: read the relevant chapter section in your notes/PowerPoints (NOT the write-up- too long), close them, do the active recall prompts here, then attempt the questions cited in the blue box.

Key vocabulary

- **Past paper** = the full official exam paper from one of the years 2020/21, 2021/22, 2022/23, 2023/24, 2024/25.
- **Assessed PST / Problem Set** = the marked problem sheets you submit every 3 weeks. There are three of them: PS1, PS2, PS3.
- **PST** = the unassessed problem sets you do every 2 weeks for tutorials. There are five: PST1–PST5.

What to focus?

Some modules are calculation and derivation heavy , such as solid state module a typical workflow is : read notes → close → write everything you remember → do questions. That works perfectly when the subject is derivation-heavy because the act of re-deriving forces structure into memory. PHAS0040 is different — it has a few short derivations but the bulk of marks come from:

- Knowing whether a reaction conserves the right quantum numbers (memorisation + a checklist)
- Drawing the correct lowest-order Feynman diagram (memorisation of vertex rules)
- Plugging numbers into a small set of named formulas (SEMF, oscillation, kinematics)
- Bookwork-style descriptions (how a detector works, what the evidence for X is, define Y)

So the same loop still works, but the 'recall' step needs to be more specific. Instead of 'write everything you remember', you need to answer targeted prompts like 'list the 6 conservation laws and the interactions in which each is conserved'. Each topic in this guide gives you those prompts.

2. The adapted revision method

This is the suggested workflow, but feel free to adopt it differently

Phase 1 — Topic-by-topic active recall

For each chapter, in this order:

- Open your lecture notes / PowerPoint slides for that topic. Read once, top to bottom, no highlighting.
- Close everything. Write out (on paper, not on screen) answers to the active recall prompts in Section 4 of this guide for that topic. Don't worry if you forget half.
- Re-open the notes and check what you got wrong. Mark gaps in red. The forgetting is the learning.
- Add anything you couldn't recall to your error notebook with an xscore (1 = small slip, 5 = total blank).

Time per chapter: 60–90 minutes. Don't spend longer — diminishing returns hit fast.

Phase 2 — Past paper questions on the same topic

Immediately after Phase 1 for a topic, do every question listed in the blue 'Where this is tested' box for that chapter. Do them closed-book with only the formula sheet (the same one given in the exam — it's at the back of the 2024 paper). Mark yourself honestly. Anything you couldn't do goes in the error notebook.

Phase 3 — Error notebook (your core asset)

Same as solid state. Every wrong answer, every 'why did I do it that way' moment, gets a one-line entry plus the correct method, with an xscore. By the end of revision this is the document you actually re-read, not the write-up.

Phase 4 — Full timed past papers

Once you've covered all high-priority topics (Phases 1+2 done for chapters 1, 2, 4, 6, 7, 8 at minimum), start full 3-hour timed papers. Use only the formula sheet. Do them in chronological order so you can see how question style has evolved. Mark and add new errors to the notebook.

Phase 5 — Final sweep

Last 3 days: stop new material. Re-read the error notebook end-to-end, twice. Re-do the highest-xscore questions one more time. Sleep.

3. Topic priority matrix

Built from counting appearances across the 5 most recent past papers (2020/21–2024/25). 'Marks/paper' is the typical mark allocation when the topic appears.

Chapter / Topic	Appears in	Marks/paper	Tier
Ch 1 — Basic ideas (4-vectors, kinematics, Yukawa, cross sections)	5/5	6–13	**A — must**
Ch 2 — Leptons & quarks (conservation laws, Feynman diagrams, neutrino oscillations)	5/5	16–22	**A — must**
Ch 3 — Experimental methods (detectors, radiation length, luminosity)	4/5	4–10	**B — high**
Ch 4 — QCD/strong (PDFs, R-ratio, colour, asymptotic freedom)	5/5	6–10	**A — must**
Ch 5 — Electroweak I (parity, V-A, charge conjugation)	3/5	2–6	**C — medium**
Ch 6 — Electroweak II (CKM, Cabibbo, kaons, CP)	4/5	6–10	**B — high**
Ch 7 — Electroweak III (Higgs, W/Z, lepton universality)	5/5	4–12	**A — must**
Ch 8 — Nuclear phenomenology (SEMF, beta/alpha decay)	5/5	8–15	**A — must**
Ch 9 — Nuclear structure (Fermi gas, shell model, spin/parity)	3/5	5–9	**B — high**
Ch 10 — Fission & fusion (energy, chain reaction, dose)	4/5	2–9	**B — high**

How to read this

- **Tier A = must master** (this is the bulk of every exam, ~40+ marks worth). Spend the most time here.
- **Tier B = high priority** (4–10 marks per paper, near-certain to appear). Do not skip.
- **Tier C = medium** (often 2–6 marks of bookwork). Worth a quick pass; don't drill.

Critical observation: the three chapters you missed (8, 9, 10) make up **between 15 and 33 marks of every single past paper**. You cannot afford to skip them — they need to be your first priority in Phase 1.

BEFORE YOU START

I also made an algorithm which predicts what is coming up this year, achieved around 60% of the accuracy when validating.

FEEL FREE TO DM ME for the PREDICTION PAPER

STOP GUESSING WHAT TO REVISE.

Start with the topics most likely to appear.

PHAS0041 SOLID STATE PHYSICS

Tiered Mock Paper and Revision Priority Guide | 2025-26 Edition

The exam is in weeks. You have a syllabus the size of a phonebook, six years of past papers, and no idea which topics deserve the next four hours of your life. Most students revise in syllabus order and run out of time before the topics that actually matter. You can do better.

WHAT WE DID

We took every PHAS0041 final exam from 2019-20 to 2024-25, hand-encoded all 97 sub-questions into a structured database, and built a quantitative model that predicts which topics tend to repeat, which tend to rotate, and which the examiner deliberately avoids in consecutive years. We tested the model the only honest way: by training it on years 1 through N and asking it to predict year N+1, blind. We did this five times.

WHAT WE FOUND

Topic-level prediction	Accuracy	What this means for you
Broad chapter (e.g. Magnetism, Diffraction)	**78.7%**	Almost four out of five major chapters in the next paper are flagged in our top tier.
Specific exercise family (e.g. powder indexing, Hund's rules workflow)	**58.6%**	Roughly six out of ten specific question types are correctly anticipated.
Anti-recency signal validated	**77%**	When a topic returns, the examiner picks a different version of it 77% of the time. We use this to predict the variant.

All accuracy figures are out-of-sample averages over 5 independent blind validation rounds. No in-sample fits. No marketing inflation.

4. Topic-by-topic deep dive

For every topic below: (a) the exact past paper, assessed problem set, and PST questions that test it (the blue box), (b) what to memorise with all key equations typeset, (c) closed-book recall prompts, (d) common traps.

Ch 1 — Basic Ideas

Tier A. Appears in every paper (~6–13 marks). Mostly kinematic calculations using four-vectors and decay-at-rest geometry.

Where this is tested

- **Past 2020/21:** Q1(b) [6m] — $D^0 \to K^-\pi^+$ decay at rest, find pion energy
- **Past 2021/22:** Q2(b) [6m] — invariant mass of multi-particle final state from 4-momenta
- **Past 2022/23:** Q1(b) [6m] — particle X decays to $\tau + \nu$, find mass of X from τ momentum
- **Past 2023/24:** Q1(a) [6m] — Z boson momentum from electron opening angle
- **Past 2024/25:** Q1(b) [6m] — B^0 decay-at-rest kinematics
- **Assessed PS1:** Q1 (natural units), Q4 (invariant mass of $\mu^+\mu^-$), Q5 (weak vs EM amplitudes), Q6 (Yukawa derivation), Q7 (branching), Q8 (cosmic ray–CMB pion threshold), Q9 (muon time dilation)
- **PST 1:** Worked solutions on E_CM (collider + fixed target), Higgs decay momentum kinematics

Key equations

Relativistic energy–momentum relation:

$$E^2 = p^2c^2 + m^2c^4$$

Four-momentum and its invariant length:

$$P = (E, cp), \quad P^2 = P\cdot P = E^2 - p^2c^2 = m^2c^4$$

Invariant mass of N particles:

$$W^2c^4 = (\Sigma\, E_i)^2 - (c\, \Sigma\, p_i)^2$$

Centre-of-mass energy:

$$\text{Colliding beams: } E_{CM}^2 = (2E_{beam})^2 = 4E_{beam}^2$$

$$\text{Fixed target: } E_{CM}^2 \approx 2E_{beam}m_{target}c^2$$

Two-particle invariant mass from energies and opening angle θ (massless limit):

$$m^2c^4 \approx 2E_1E_2\,(1 - \cos\theta)$$

Yukawa potential and force range:

$$V(r) = -(g^2/4\pi)\, e^{-r/R} / r, \quad R = \hbar / (M_X c)$$

Number of events from luminosity and cross section:

$$N = \sigma \times \int L\, dt = \sigma \times \mathcal{L}$$

Conversion factor (memorise):

$$\hbar c = 0.197 \text{ GeV}\cdot\text{fm}$$

Closed-book recall prompts

- Write the relativistic energy–momentum relation. Use it to express momentum in terms of E and m.
- A particle of mass M decays at rest into two equal-mass daughters of mass m. Write the energy of each daughter.

- Two photons are observed with energies E_1 and E_2 with an angle θ between them. Write the invariant mass of the parent.
- Derive $R = \hbar/(Mc)$ from the uncertainty principle for an exchanged virtual particle of mass M.
- Convert 125 GeV/c² into kg. Convert a length of 23 GeV^{-1} into metres. (PS1 Q1 — exact)

Common traps

- Forgetting that 'decay at rest' means total momentum = 0 in the lab frame, so daughter momenta are opposite.
- Mixing up E_CM^2 for fixed target ($\approx 2Em\,c^2$) with that for colliding beams ($= 4E^2$).
- Forgetting factors of c when working in SI; better to work in natural units throughout and restore c at the end.

Ch 2 — Leptons and Quarks

Tier A. The single biggest topic in the exam. Q1 of every paper (14 marks) is the same question type: 'which of these reactions are allowed and which forbidden, draw Feynman diagrams for the allowed ones'.

Where this is tested

- **Past 2020/21:** Q1(a) [14m] — allowed/forbidden + Feynman diagrams (Λ^0, ψ^*, ve, μ^+e^- reactions)
- **Past 2021/22:** Q2(a) [14m] — allowed/forbidden + Feynman; Q1(a) [8m] — neutrino oscillations, $\sin^2(2\theta)$; Q1(c) [6m] — infer B, L, quark content of unknown particle
- **Past 2022/23:** Q1(a) [14m] — allowed/forbidden; Q2(a) [10m] — neutrino oscillations $\sin^2(2\theta)$, Δm^2, survival probability
- **Past 2023/24:** Q1(b) [14m] — allowed/forbidden; Q2(b) [6m] — neutrino survival probability
- **Past 2024/25:** Q1(a) [14m] — allowed/forbidden + Feynman diagrams
- **Assessed PS1:** Q2 (allowed processes including $\mu^+ \to e^+\nu_e\bar{\nu}_\mu$), Q3 (Feynman diagrams $\mu^-\mu^-$ scattering), Q10 (allowed reactions including $\nu_\tau p \to \tau^-\Delta^{++}$)
- **PST 3:** Tutorial-style allowed/forbidden problems

The conservation law table — memorise this in 60 seconds

Quantity	Strong	EM	Weak	Notes
Energy / momentum	✓	✓	✓	Always conserved
Electric charge Q	✓	✓	✓	Always conserved
Baryon number B	✓	✓	✓	$B = N(q)/3 - N(\bar{q})/3$
Lepton number L (total)	✓	✓	✓	$L = N(\ell^-) - N(\ell^+) + N(\nu) - N(\bar{\nu})$
Individual L_e, L_μ, L_τ	✓	✓	✓	Yes, except in neutrino oscillation
Quark flavour (u,d,s,c,b,t)	✓	✓	✗	ONLY weak changes flavour
Parity P	✓	✓	✗	Wu experiment 1957
Charge conjugation C	✓	✓	✗	
CP	✓	✓	≈	Tiny violation in K, B systems

Key equations

Lepton number (total and per generation):

$$L = N(\ell^-) - N(\ell^+) + N(\nu) - N(\bar{\nu})$$

Baryon number:

$$B = N_q / 3$$

Sargent's rule for charged lepton lifetime:

$$1 / \tau_\ell \propto m_\ell^5$$

Two-flavour neutrino oscillation probability:

$$P(\nu_\alpha \to \nu_\beta) = \sin^2(2\theta)\ \sin^2[\ 1.27\ \Delta m^2[\mathrm{eV}^2] \cdot L[\mathrm{km}] / E[\mathrm{GeV}]\ ]$$

Survival probability of the original flavour:

$$P(\nu_\alpha \to \nu_\alpha) = 1 - P(\nu_\alpha \to \nu_\beta)$$

The 'is this reaction allowed?' checklist

Apply in this order to every reaction:

- 1. Electric charge conserved? (Sum Q_initial = Sum Q_final)
- 2. Energy: is the total mass of the products less than the parent mass (for decays) or less than the CM energy (for scatterings)?
- 3. Baryon number conserved? (Count quarks − antiquarks, divide by 3)
- 4. Total lepton number conserved? (Count leptons − antileptons across all generations)
- 5. Individual lepton numbers L_e, L_μ, L_τ each conserved? (Most common trap — students forget this and only check total L)
- 6. If only strong/EM possible (no leptons, short timescale): is quark flavour conserved?
- 7. If forbidden, name the rule. If allowed, draw the lowest-order Feynman diagram and state which interaction (strong/EM/weak).

Feynman diagram building blocks

- Photon couples only to electric charge. Vertices: $e^-e^-\gamma$, $qq\gamma$. Photon never couples to neutrinos.
- Gluon couples only to colour (i.e. quarks). qqg vertex. Gluons also self-couple (ggg, gggg vertices).
- W± couples leptons within a generation (e.g. $e^-\bar{\nu}_e W^-$) and quarks across generations with CKM weights (u→d strong, u→s weaker, u→b very weak).
- Z^0 couples to all fermions but does NOT change flavour. So no FCNC at tree level.
- Higgs couples to mass — heavier particles have stronger Higgs coupling.

Lepton properties to memorise

- Three generations: (e, ν_e), (μ, ν_μ), (τ, ν_τ). Charged lepton mass increases by ~×200 each generation.
- Sargent's rule: lepton lifetime $\tau_\ell \propto m_\ell^{-5}$. So $\tau_\tau << \tau_\mu << \tau_e$ (electron stable).
- Branching fraction $Br(\tau \to e\, \nu_\tau\, \bar{\nu}_e) \approx 1/5$ because τ has 5 channels: $\to e\bar{\nu}\nu$, $\to \mu\bar{\nu}\nu$, $\to u\bar{d}\nu$ (×3 colours).

Neutrino oscillation tricks

- 'Maximum oscillation' means $\sin^2[\ldots] = 1$, so the bracket = π/2. This pins down Δm^2 when L_max is given.
- 'Amplitude of oscillation' = $\sin^2(2\theta)$, found from (max number of oscillated neutrinos) / (original number).
- Implication: neutrinos must have mass (otherwise $\Delta m^2 = 0$ and there'd be no oscillation). Breaks SM prediction of massless neutrinos.

Closed-book recall prompts

- Write the conservation law table from memory in 60 seconds.
- Without looking, list the 6 quark flavours, their charges, and approximate masses.
- Sketch the Feynman vertices for: γqq, gqq, $W^-e\bar{\nu}_e$, $Z\mu^+\mu^-$.
- Write the neutrino oscillation formula. Explain physically why oscillation requires neutrinos to have mass.
- Reaction $\pi^0 \to e^+e^-e^+e^-$: allowed or forbidden? (Allowed via two virtual photons.)
- Reaction $\mu^- \to e^- + \gamma$: allowed or forbidden? (Forbidden — violates L_e and L_μ individually.)

Common traps

- Forgetting individual lepton number conservation — checking only total L.
- Drawing a Z exchange that changes flavour. Z does NOT change flavour.
- Forgetting that the photon does not couple to neutrinos.

Ch 3 — Experimental Methods

Tier B. Appears in 4/5 papers, usually as a 4–10 mark sub-question on either how a detector measures particles, or a calculation involving radiation length, luminosity, or Cherenkov.

Where this is tested

- **Past 2020/21:** Q3(b)(ii) [5m] — describe how an HW → $b\bar{b}\mu\nu$ event is recorded in ATLAS; Q3(c) [2m] — DIS energy needed to resolve 10^{-18} m
- **Past 2021/22:** Q3(a) [9m] — heavy particle X discovery signatures (which final state, mass measurement)
- **Past 2023/24:** Q2(a) [4m] — radiation length from electron stopping
- **Past 2024/25:** Q2(a)(iii) [2m] — jet measurement at LHC
- **Assessed PS2:** Q7 — for X → ZZ → $e^+e^-\mu^+\mu^-$, what is seen in tracker, EM cal, hadronic cal, muon chambers

Key equations

Bending radius of a charged particle in a magnetic field:

$$R[\mathrm{m}] = 3.3 \times p[\mathrm{GeV}] / B[\mathrm{T}]$$

Calorimeter energy resolution scaling:

$$\sigma(E) / E \propto 1 / \sqrt{E}$$

Instantaneous luminosity (circular collider):

$$\mathcal{L} = nN_1N_2f / A$$

Radiation length scaling:

$$1 / X_0 \propto Z^2 / A$$

Maximum number of particles in an EM shower:

$$N_{\mathrm{max}} = E_0 / E_c = 2^{t_max}$$

Cherenkov cone opening angle:

$$\cos \theta_C = 1 / (n\beta)$$

What to memorise

- Multi-purpose detector shell structure (inside → out): vertex/tracking detectors (in B-field) → EM calorimeter → hadronic calorimeter → muon chambers
- How each particle is identified: photon = EM cal only (no track); electron = track + EM cal; muon = track + reaches muon chambers; charged hadron = track + hadronic cal; neutral hadron = hadronic cal only; neutrino = missing transverse momentum; b-jet = displaced secondary vertex
- Bethe–Bloch (ionisation energy loss): proportional to z^2/v^2. Used for particle ID at moderate momenta.
- Photon interactions vs energy: photoelectric ($\propto Z^5/E$^3.5) at low E, Compton ($\propto Z \ln E/E$) at intermediate E, pair production ($\propto Z^2$) above 2m_e.
- Calorimeters get RELATIVELY more accurate at high energy; trackers get worse.

Closed-book recall prompts

- Sketch the cross section of a generic collider detector and label every layer.
- For each of {photon, electron, muon, charged pion, neutron, neutrino, b-jet}, describe what it leaves in each layer.
- Write the radiation length scaling and explain why high-Z materials (lead, tungsten) make good EM calorimeters.

- A 10 GeV electron stops in 2 m of material at critical energy 10 MeV. Estimate X_0 of the material. (2023/24 Q2(a) — exact.)
- Cherenkov: write the threshold β for emission.

Common traps

- Confusing X_0 (EM, electrons/photons) with nuclear absorption length λ_a (hadronic, hadrons). Different Z dependence.
- Forgetting that the muon is the only charged particle that traverses the whole detector.
- Forgetting that 'missing momentum' is reconstructed in the transverse plane only.

Ch 4 — QCD / Strong Interactions

Tier A. Appears in 5/5 papers. Usually a calculation using the R-ratio, or LHC parton kinematics (x_1, x_2).

Where this is tested

- **Past 2020/21:** Q2(b) [4m] — branching ratio Br(Φ → uū) with colour factor; Q2(c) [4m] — gluon fusion x_1, x_2 at LHC for Y at 260 GeV
- **Past 2021/22:** Q1(b) [6m] — relative change in σ(e^+e^- → anything) and R from a new heavy lepton
- **Past 2024/25:** Q2(a) [10m] — LHC parton kinematics, jet opening angle, jet measurement
- **Assessed PS2:** Q1 (α_s running for Q = 10 GeV and 4000 GeV; with extra quark flavours), Q5 (Higgs LHC parton kinematics x_1x_2)

Key equations

R-ratio (with colour factor):

$$R = \sigma(e^-e^+ \to q\,\bar{q}) / \sigma(e^-e^+ \to \mu^+\mu^-) = N_C\, \Sigma\, e_q^2$$

With N_C = 3 colours and 5 accessible quark flavours (u,d,s,c,b):

$$R = 3 \times (4/9 + 1/9 + 1/9 + 4/9 + 1/9) = 11/3$$

Running of α_s with momentum transfer:

$$\alpha_s(Q^2) = \alpha_s(\mu^2) / [\, 1 + B\, \alpha_s(\mu^2) \ln(Q^2/\mu^2)\,]$$

$$B = (33 - 2N_f) / (12\pi)$$

Confining potential between coloured objects:

$$V(r) = -\, \alpha_s \hbar c / r + \lambda\, r, \quad \lambda \approx 1 \text{ GeV/fm}$$

DIS four-momentum transfer (deep inelastic scattering):

$$q^2 = (k - k')^2 \approx 2\, EE' (\cos\theta - 1)$$

Bjorken-x parton momentum fraction:

$$x = -\, q^2 / (2\, p \cdot q)$$

LHC cross section (parton convolution):

$$\sigma(pp \to X) = \Sigma \iint f_i(x_1, Q^2)\, f_j(x_2, Q^2)\, \sigma_{ij\to X}\, dx_1\, dx_2$$

Threshold condition to produce a particle of mass M at √s:

$$x_1 x_2 \geq M^2 / s$$

If $x_1 = x_2$ (symmetric / particle at rest in CM):

$$x = M / \sqrt{s}$$

What to memorise

- Three colours (red, green, blue). Quarks carry colour, antiquarks anticolour. Hadrons are colour singlets.
- Asymptotic freedom: α_s DECREASES at high Q^2 (short distance). Confinement: $\alpha_s \to \infty$ at large r. Cause: gluon self-coupling.
- For N_f < 16.5, B > 0 and α_s runs down with Q^2 (asymptotic freedom).
- To resolve structure of size d, need $q^2 \gtrsim (\hbar c/d)^2$. Higher energy → finer resolution.
- At low x and high Q^2, gluons and sea quarks dominate.

- Why $p\bar{p}$ collisions: at high x where valence quarks dominate, p has uud and $\bar{p}$ has $\bar{u}\bar{u}\bar{d}$ → annihilation cross section is huge. At LHC (low x) this advantage disappears.

Closed-book recall prompts

- Write the R-ratio formula. Predict R above the b-quark threshold (5 quark flavours).
- Sketch $\alpha_s(Q^2)$ and explain 'asymptotic freedom' in one sentence.
- A new particle of mass 200 GeV is produced at $\sqrt{s}$ = 13 TeV with $x_1 = 2\,x_2$. Find x_2. (PS2 Q5 — exact)
- Why are $p\bar{p}$ collisions historically used to discover heavy particles (e.g. W/Z)?
- Sketch the Feynman diagram for $e^+e^- \to q\,\bar{q}\,g$ (gluon emission).

Common traps

- Forgetting the colour factor of 3 in the R-ratio.
- Including a quark in the R-ratio sum that is kinematically inaccessible (e.g. top quark at $\sqrt{s}$ = 30 GeV).
- Mixing up Q^2 and q^2 conventions — they differ by a sign.

Ch 5 — Electroweak Interactions: Part I

Tier C. Lower priority — typically 2–6 marks of bookwork on parity violation or the V-A structure.

Where this is tested

- **Past 2024/25:** Q2(c) [2m] — parity violation evidence (bookwork)
- **Assessed PS2:** Q2 (C-parity of π^0 from $\pi^0 \to \gamma\gamma$; what does $\pi^0 \to \gamma\gamma\gamma$ being forbidden imply about C-parity of γ); Q4 (which of $\nu\mu + \mu^-$ vs $\nu e + \mu^-$ is unambiguous evidence for weak neutral currents)

What to memorise

- Parity (P): the spatial inversion operator. $P(x) = -x$. The weak interaction violates parity: it treats left- and right-handed particles differently. Strong and EM conserve P.
- Wu experiment (1957): observed asymmetry in β-decay of polarised ^{60}Co — electrons preferentially emitted opposite to the nuclear spin. Smoking gun for parity violation.
- Charge conjugation (C): swaps a particle for its antiparticle. The weak interaction violates C and very weakly violates CP (observed in K and B mesons).
- V−A structure: the W boson only couples to LEFT-handed particles and RIGHT-handed antiparticles. This is why right-handed neutrinos are not observed.
- Helicity suppression in pion decay: $\pi^+ \to e^+\nu_e$ is much rarer than $\pi^+ \to \mu^+\nu_\mu$, suppressed by $\sim(m_e/E)^2$, because V−A forces the e^+ to have the 'wrong' helicity.

Closed-book recall prompts

- State one experimental piece of evidence for parity violation. (Wu 1957 ^{60}Co experiment.)
- Why does the weak interaction not allow right-handed neutrinos?
- Why is $\pi^+ \to e^+\nu_e$ suppressed relative to $\pi^+ \to \mu^+\nu_\mu$ despite more phase space?
- If $\pi^0 \to \gamma\gamma$ is allowed, what is the C-parity of π^0? (PS2 Q2 — answer: $C(\pi^0) = +1$, since $C(\gamma) = -1$ and there are 2 photons.)

Ch 6 — Electroweak: Part II (Quark Mixing, CKM)

Tier B. Appears in 4/5 papers via Feynman diagrams of quark-level decays. The skill is recognising Cabibbo-favoured vs suppressed transitions.

Where this is tested

- **Past 2022/23:** Q3(a) [6m] — Feynman diagrams for $D^+ \to \bar{K}^0 e^+ \nu_e$, $D^0 \to K^- \pi^+$, $D^0 \to K^+ \pi^-$; Q3(b) [2m] — explain why $D^0 \to K^+ \pi^-$ rate << $D^0 \to K^- \pi^+$; Q3(c) [2m] — estimate the ratio
- **Past 2024/25:** Q2(b) [8m] — B meson decays + identify interaction type + decay rate ordering
- **Assessed PS2:** Q3 — Feynman diagram for $B^+ \to J/\psi\, K^{*+}$ and subsequent $K^{*+} \to \pi^0 K^+$
- **PST 4:** Q1 — Feynman diagrams for $B^0 \to D^- \pi^+$, $B^0 \to \pi^+ \pi^-$, $B^0 \to J/\psi\, K^0$; order by decreasing decay rate

Key equations / data

CKM quark-mixing matrix (memorise the rough magnitudes):

$$V_{CKM} \approx \begin{vmatrix} 0.974 & 0.225 & 0.004 \\ 0.225 & 0.973 & 0.042 \\ 0.009 & 0.041 & 0.999 \end{vmatrix}$$

Weak mixing angle:

$$\cos \theta_W = M_W / M_Z$$

What to memorise

- CKM rough magnitudes: $|V_{ud}| \approx 0.97$, $|V_{us}| \approx 0.22$, $|V_{cb}| \approx 0.04$, $|V_{ub}| \approx 0.004$. Diagonal large, off-diagonal small. Further from the diagonal → more suppressed.
- Cabibbo-favoured: u↔d, c↔s, t↔b (large CKM). Cabibbo-suppressed: u↔s, c↔d (small). Doubly-Cabibbo-suppressed: c↔b in some senses; $D^0 \to K^+ \pi^-$ requires both c→d AND u→s.
- Worked example: $D^0(c\bar{u}) \to K^-(s\bar{u})\pi^+(u\bar{d})$ is Cabibbo-favoured (c→s). $D^0(c\bar{u}) \to K^+(u\bar{s})\pi^-(d\bar{u})$ is doubly Cabibbo-suppressed (c→d AND u→s) → rate $\sim |V_{us}|^4 \approx 0.0025$ of the favoured.
- Z boson does NOT mix flavours. No flavour-changing neutral currents at tree level (the GIM mechanism).
- Kaon system: K^0 and $\bar{K}^0$ mix via box diagrams (W exchange). Generates K_S (CP-even, short-lived) and K_L (CP-odd, long-lived). Tiny CP violation observed in K_L decays.

How to draw weak quark decay Feynman diagrams

- 1. Identify the spectator quarks (those that don't change in the decay).
- 2. Identify the active quark transition (e.g. b→c, c→s, s→u).
- 3. The active quark emits a W boson which decays into leptons ($W \to \ell \bar{\nu}$) or another quark pair ($W \to q\, \bar{q}'$).
- 4. Spectators sit on horizontal lines untouched.
- 5. Check that initial and final hadrons have the right quark content.

Closed-book recall prompts

- Reproduce the rough magnitudes of $|V_{ud}|$, $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$.
- Order these decays by rate (fastest → slowest): $B^0 \to D^- \pi^+$, $B^0 \to \pi^+ \pi^-$, $B^0 \to J/\psi\, K^0$. (PST4 Q1 — exact.)
- Draw the lowest-order Feynman diagram for $B^-(\bar{b}u) \to D^0(c\bar{u}) + \pi^-(\bar{d}u)$.
- Why is $K^0 \to \mu^+ \mu^-$ extremely rare? (No FCNC; only via box diagram → very suppressed.)

Ch 7 — Electroweak: Part III (Higgs, W/Z, unification)

Tier A. Appears in all 5 papers. Usually one calculation about Higgs production/decay rates plus some bookwork on lepton universality or W/Z discovery.

Where this is tested

- **Past 2020/21:** Q2(a) [5m] — H → XX kinematics and X flight distance from natural width; Q3(b) [10m] — HW → $b\bar{b}\mu\nu$ production count + ATLAS detection + b-jet kinematics
- **Past 2021/22:** Q3(a) [9m] — heavy X = 250 GeV decaying to ZZ/WW: discovery channels, mass measurement strategies
- **Past 2022/23:** Q3(d) [2m] — Higgs at LEP, max √s; Q3(e) [2m] — gluon-gluon fusion; Q3(f) [2m] — dominant Higgs decay; Q3(g) [4m] — H → ZZ event count
- **Past 2023/24:** Q3(a) [13m] — $e^+e^- \to \mu^+\mu^-$ cross section + A_FB derivation; Q3(b) [7m] — W decay branching via lepton universality
- **Assessed PS2:** Q5 (Higgs production at LHC, x_1x_2, gluon dominance, $H \to \mu^+\mu^-$ rarity); Q6 (why $H \to b\bar{b}$ took so long; $H \to c\bar{c}$? $H \to t\bar{t}$?)

Key equations

Weak mixing angle:

$$\cos \theta_W = M_W / M_Z$$

Number of events from cross section, luminosity, branching ratio:

$$N = \sigma \times \mathcal{L} \times \mathrm{Br}$$

Forward–backward asymmetry (2023/24 Q3(a)):

$$A_{FB} = (\sigma_F - \sigma_B) / (\sigma_F + \sigma_B)$$

Total cross section from a differential one (2023/24 Q3(a)(ii)):

$$\sigma_{total} = \int (d\sigma/d\Omega)\, d\Omega$$

What to memorise

- EW unification: at high energy ($q^2 >> M_W^2$) the weak and EM interactions have similar strength. They differ at low energy because of the W mass.
- Discovery of neutral currents (1973, Gargamelle bubble chamber, CERN): $\nu_\mu N \to \nu_\mu X$ — neutrino scatters off matter without changing identity. Required a neutral exchange particle.
- Discovery of W and Z (1983, UA1/UA2 at SppS, CERN): produced in $p\bar{p}$ via $q\bar{q} \to W/Z$. $W \to e\nu$ (lepton + missing pT). $Z \to ee$ or $\mu\mu$ (two leptons reconstructing M_Z).
- Higgs mechanism: gives mass to W, Z and fermions via spontaneous symmetry breaking. Particles couple with strength proportional to their mass.
- Higgs production at LHC: dominant mode is gluon-gluon fusion ($gg \to H$ via top quark loop). Other modes: VBF, VH (associated with W/Z), $t\bar{t}H$.

Higgs branching ratios at m_H = 125 GeV (rough, memorise the order):

- $b\bar{b}$ ~58% (dominant)
- WW ~21%
- gg ~9%
- ττ ~6%
- $c\bar{c}$ ~3%
- ZZ ~2.7%
- γγ ~0.2%
- μμ ~0.02%

- Discovery channels (Higgs 2012): $H \to \gamma\gamma$ (rare but clean — narrow peak in two-photon mass) and $H \to ZZ \to 4\ell$ (rare but very clean — four leptons reconstruct m_H).
- Why $H \to b\bar{b}$ took so long: enormous QCD background of $b\bar{b}$ pairs from non-Higgs processes.
- H does NOT decay to $t\bar{t}$: m_t (173 GeV) > m_H (125 GeV) / 2.

Lepton universality and W decays

- W couples equally to all three lepton generations and to all kinematically allowed up–down quark pairs.
- W^- decay channels (ignoring CKM suppression): $e^-\bar{\nu}_e$, $\mu^-\bar{\nu}_\mu$, $\tau^-\bar{\nu}_\tau$, $d\bar{u}$, $s\bar{c}$. The $b\bar{t}$ channel is forbidden ($m_t > m_W$).
- Quark channels each get colour factor 3, so total weight = 1 + 1 + 1 + 3 + 3 = 11.
- $Br(W \to e\nu_e) \approx 1/11 \approx 9\%$ (sometimes simplified to ~1/9 = 11%, the exam may accept either).

Closed-book recall prompts

- Sketch the dominant Higgs production mechanism at the LHC and explain why it's dominant.
- What is the dominant Higgs decay mode? Why was it not the discovery channel? (PS2 Q6 — exact)
- Explain how the W boson was discovered — what experimental signature was used?
- Estimate $Br(W^+ \to e^+\nu_e)$. Hence estimate $Br(W^+W^- \to e^+e^-\nu_e\bar{\nu}_e)$. (2023/24 Q3(b) — exact)
- Calculate the number of $H \to ZZ \to 4\ell$ events with $\sigma_H = 50$ pb, $\int L\,dt = 50\ fb^{-1}$, $Br(H \to ZZ) = 2.7\%$, $Br(Z \to ee \text{ or } \mu\mu) = 6.8\%$. (2022/23 Q3(g) — exact)

Common traps

- Forgetting the colour factor of 3 for quark channels in W branching ratios.
- Saying $H \to t\bar{t}$ is allowed (it isn't — $m_H < 2m_t$).
- Confusing $H \to \gamma\gamma$ as 'common' — it's only ~0.2%, but it's clean.

Ch 8 — Nuclear Phenomenology

Tier A — and one of your missed lectures. 5/5 papers. The SEMF appears in some form in every paper, often the most heavily-weighted single sub-question (8–14 marks).

Where this is tested

- **Past 2020/21:** Q3(a) [8m] — use SEMF to determine if ^{75}Ge undergoes β^+ or β^- decay and energy released
- **Past 2021/22:** Q3(b) [9m] — show lowest mass nucleus with A=64 is $^{64}_{28}$Co; Q3(c) [2m] — change in nuclear surface when A doubles ×8
- **Past 2022/23:** Q2(b) [3m] — ^{14}C decays per gram per minute; Q2(c) [2m] — binding energy of ^{2}H from fusion energy release
- **Past 2023/24:** Q2(c) [6m] — compare BE of ^{64}Zn from atomic mass and SEMF; Q2(d) [4m] — ^{14}C dating
- **Past 2024/25:** Q3(a) [11m] — bookwork: define isotope, β^+ decay, evidence for shell model; Q3(b) [9m] — proton decay lifetime from radiation dose limit
- **Assessed PS3:** Q1 (which SEMF terms increase/decrease BE and why); Q2 (most stable Z for A=145)
- **PST 5:** Q2 — derive Z_min ≈ a_a A / (2 a_a + 2 a_C A^(2/3))

Key equations

The SEMF (semi-empirical mass formula) — write it on a card and stick it on your wall:

$$M(Z, A) = Z(m_p + m_e) + (A - Z)\, m_n$$
$$- a_v A + a_s A^{2/3} + a_C Z^2 / A^{1/3} + a_a (Z - A/2)^2 / A \pm \delta\, a_p A^{-1/2}$$

Constants (memorise these):

$$a_v = 15.67, \quad a_s = 17.23, \quad a_C = 0.714, \quad a_a = 93.15, \quad a_p = 11.1 \quad \text{(all MeV)}$$

Most stable Z for given A (PST5 Q2 — derive this by setting $\partial M/\partial Z = 0$):

$$Z_{min} \approx a_a A / [\, 2\, a_a + 2\, a_C A^{2/3} \,]$$

Radioactive decay law:

$$N(t) = N_0\, e^{-t/\tau}$$

Activity (decay rate):

$$A = -\, dN/dt = N / \tau$$

Half-life vs mean life:

$$T_{½} = \tau \ln 2$$

Radiation dose (used in 2024/25 Q3(b) proton decay calculation):

$$1 \text{ Gy} = 1 \text{ J/kg}$$

SEMF term-by-term

Term	Sign	Physical origin	Effect on stability
Volume −a_v A	−	Each nucleon binds to its neighbours; bulk binding energy	Strongly bound, ∝ A
Surface +a_s A^(2/3)	+	Surface nucleons have fewer neighbours, so less bound	Reduces binding; matters for small A
Coulomb +a_C Z²/A^(1/3)	+	Protons repel each other electrostatically	Reduces binding for high Z; favours fewer protons

Term	Sign	Physical origin	Effect on stability
Asymmetry +a_a (Z−A/2)²/A	+	Pauli exclusion (Fermi gas explanation): adding extra protons OR neutrons costs kinetic energy	Most stable when Z = N. For heavy nuclei, Coulomb pushes towards N > Z.
Pairing ±δ a_p A^(−1/2)	± / 0	Like-spin pairing: even-even gets extra binding, odd-odd gets penalty, odd-A is zero	Favours even-even nuclei

Beta decay rules

- β^- decay: n → p + e^- + $\bar{\nu}$_e. Z → Z+1, A unchanged. When parent has too many neutrons.
- β^+ decay: p → n + e^+ + ν_e. Z → Z−1, A unchanged. When parent has too many protons. Q-value must exceed 2 m_e c² ≈ 1.022 MeV.
- Electron capture: p + e^- → n + ν_e. Same effect as β^+ but no positron. Often competes with β^+.
- α decay: nucleus emits ^{4}He. Z → Z−2, A → A−4. Common for very heavy nuclei (A > 200).
- γ decay: nucleus de-excites by emitting a photon. No change in Z or A.
- For fixed A, the SEMF is a parabola in Z. Min = most stable isobar. Right of min: β^+ or EC. Left of min: β^-.
- Even-A gives TWO parabolas (even-even below odd-odd, separated by 2δ). Odd-A gives a single parabola.

Closed-book recall prompts

- Write the SEMF from memory (the structure, not the constants). Name each term in physical words.
- Why does the asymmetry term favour Z = N? (Fermi gas — adding extra of one kind costs Fermi kinetic energy.)
- Why do heavy nuclei have N > Z? (Coulomb term grows as Z², so it pays to have neutrons rather than more protons.)
- Derive Z_min for fixed A by ∂M/∂Z = 0. (PST5 Q2 — exact.)
- Carbon dating: a sample shows 3.5 decays/min vs 12.4 for fresh wood, half-life 5730 years. How old? (2023/24 Q2(d) — exact.)
- Use SEMF to compute the BE of ^{64}Zn from its atomic mass 63.929 amu, compare to SEMF prediction. (2023/24 Q2(c) — exact.)
- Determine if ^{75}Ge undergoes β^+ or β^-. (2020/21 Q3(a) — exact.)

Common traps

- Forgetting to include electron masses when computing atomic vs nuclear masses (the SEMF as written includes m_e in the (m_p + m_e) term — it gives ATOMIC mass).
- Forgetting that β^+ has a 1.022 MeV threshold (need 2 m_e c² of available energy).
- Mixing up odd-A and even-A parabolas: even-A has two.

Ch 9 — Nuclear Structure (Fermi Gas + Shell Model)

Tier B — your missed lecture. Appears in 3/5 papers, usually as a 5–9 mark question on shell model spin/parity prediction. The 2024 paper asked for 'evidence for the shell model' (bookwork) — a regular type.

Where this is tested

- **Past 2020/21:** Q2(e) [5m] — given ^{73}Ge (J=9/2, +) and ^{73}Ga (J=3/2, −), predict spin and parity of ^{72}Ga
- **Past 2024/25:** Q3(a)(ii) [2m] — bookwork: evidence for shell model
- **Assessed PS3:** Q4 — first excited state of ^{9}Be assuming only neutrons excited; Q5 — shell model configurations for ^{40}Ar, ^{39}K, ^{40}Ca and predict J^P; Q6 — find j and ℓ given a shell holds 16 nucleons

Key equations

Density of states in a 3D box:

$$n(p)\,\mathrm{d}p = (4\pi\, V) / (2\pi\, \hbar)^3 \cdot p^2\, \mathrm{d}p$$

Number of proton / neutron states in the Fermi gas:

$$Z = V\,(p_F^{\,p})^3 / (3\pi^2\, \hbar^3), \quad N = V\,(p_F^{\,n})^3 / (3\pi^2\, \hbar^3)$$

Fermi momentum (assuming Z = N = A/2):

$$p_F = (\hbar / R_0)\,(9\pi/8)^{1/3}$$

Fermi energy (numerically ~33 MeV):

$$E_F = p_F^{\,2} / (2\,M)$$

Average kinetic energy per nucleon:

$$\langle E_{kin}\rangle = (3/5)\,E_F$$

Total kinetic energy of the nucleus:

$$E_{kin}(N, Z) = (3/10M) \times [N\,(p_F^{\,n})^2 + Z\,(p_F^{\,p})^2]$$

Woods–Saxon potential + spin-orbit (the shell model potential):

$$V(r) = -\,V_0 / (1 + \exp((r - a)/b)) + V_{\ell s}(r)\,\langle \mathrm{L}\cdot\mathrm{S}\rangle / \hbar^2$$

Magic numbers (memorise)

2, 8, 20, 28, 50, 82, 126

Energy ordering of shells (memorise the first ~10)

Filled in this order (cumulative occupations give the magic numbers):

1s_(1/2), 1p_(3/2), 1p_(1/2), 1d_(5/2), 2s_(1/2), 1d_(3/2),

1f_(7/2), 2p_(3/2), 1f_(5/2), 2p_(1/2), 1g_(9/2), …

Predicting spin and parity J^P

- Step 1: Fill shells (separately for protons and neutrons) up to Z and N. Max 2j+1 nucleons of each type per (n, ℓ, j) shell.
- Step 2: Pairing — within a shell, like nucleons pair to give J = 0. Filled shells contribute nothing to total J.
- Step 3: Even-even nucleus → all nucleons paired → J^P = 0^+. Always.
- Step 4: Odd-A (one unpaired nucleon) → J = j of unpaired nucleon, P = (−1)^ℓ.
- Step 5: Odd-odd → J between |j_p − j_n| and j_p + j_n; P = (−1)^(ℓ_p + ℓ_n). Often ambiguous — use neighbouring nuclei (as in 2020/21 Q2(e)).

Closed-book recall prompts

- List the magic numbers.
- Without spin-orbit coupling, what magic numbers does the shell model predict? (Wrong: 2, 8, 20, 40, 70, 112.)
- Predict J^P of ^{15}O (Z=8, N=7), ^{17}O (Z=8, N=9), ^{40}Ca (Z=N=20).
- Why are even-even nuclei always 0^+ in the ground state?
- Estimate the Fermi energy of nucleons in a heavy nucleus. Compare to typical binding energies.
- Give two pieces of experimental evidence for the existence of nuclear shells. (2024/25 Q3(a)(ii) — exact.)
- Configurations of ^{40}Ar, ^{39}K, ^{40}Ca? Is ^{40}Ca expected to be stable? Why? (PS3 Q5 — exact.)

Common traps

- Forgetting that protons and neutrons fill INDEPENDENT shells.
- Forgetting that spin-orbit pushes $j = \ell + ½$ DOWN, $j = \ell − ½$ UP (opposite sign convention from atomic physics).
- For odd-odd, expecting a unique answer when there isn't one.

Ch 10 — Fission and Fusion

Tier B — your missed lecture. Appears in 4/5 papers — typically a small calculation (energy released in fission/fusion, chain reaction, or a dose calculation).

Where this is tested

- **Past 2020/21:** Q2(d) [2m] — energy released in $^{3}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + 2p$ from given binding energies
- **Past 2022/23:** Q2(c) [2m] — BE of $^{2}_{1}H$ from $p + p \rightarrow ^{2}_{1}H + e^{+} + \nu_e + 0.42$ MeV; Q2(d) [5m] — derive $E_f/E_i = ((A-1)/(A+1))^2$ for elastic n + nucleus(A) collision
- **Past 2024/25:** Q3(b) [9m] — proton decay lifetime from radiation dose limit (each proton decay → $\sim m_p c^2 \approx 1$ GeV deposited, find τ such that dose < 5 Gy/year)
- **Assessed PS3:** Q3 — up to which element is fusion energetically favourable? Explain.
- **PST 5:** Q1 — Sun lifetime: given $N = 9\times10^{56}$ initial H, luminosity 3.86×10^{26} W, PPI chain energy 26.7 MeV minus 0.26 MeV per neutrino × 2, find remaining time.

Key equations

Chain reaction in a fissile material (n = neutrons per fission, q = probability of inducing another fission, t_p = neutron lifetime):

$$N(t) = N(0)\exp[\,(nq - 1)\,t / t_p\,]$$

- Critical: = 1. Supercritical: (bomb). Subcritical: .

Coulomb barrier for fusion of two nuclei (Z, R) and (Z′, R′):

$$V_C = (e^2 / 4\pi\varepsilon_0)\cdot ZZ' / (R + R')$$

Elastic head-on collision: ratio of neutron energy after to before collision with a nucleus of mass number A (2022/23 Q2(d)):

$$E_f / E_i = ((A - 1) / (A + 1))^2$$

Spontaneous fission criterion (where Coulomb wins over surface):

$$Z^2 / A \geq 49 \quad (\text{i.e. } Z \gtrsim 116)$$

PP-I chain net reaction in the Sun (memorise the energy):

$$4\,p \rightarrow {}^4He + 2\,e^+ + 2\,\nu_e + 26.7\text{ MeV}$$

What to memorise

- Binding energy per nucleon peaks around A ≈ 56 (iron). Below: fusion releases energy. Above: fission releases energy.
- Energy released per fission of $^{235}U \approx 200$ MeV. Compute by SEMF differences before/after.
- Induced fission: $^{235}U + n \rightarrow ^{236}U^* \rightarrow$ fission products + ~2.5 neutrons.
- Why moderation matters: thermal neutrons (~0.025 eV) have larger fission cross sections on ^{235}U than fast neutrons. Light nuclei (small A) make good moderators because the elastic energy ratio $((A-1)/(A+1))^2$ is smallest for them — they remove the most energy per collision.
- Coulomb barrier for two protons ~1 MeV; thermal energy at 10^7 K is ~1 keV → fusion in stars proceeds by quantum tunnelling.
- PP-I chain: $p+p \rightarrow {}^2H + e^+ + \nu_e$ (slow, weak interaction limits the rate); ${}^2H + p \rightarrow {}^3He + \gamma$; ${}^3He + {}^3He \rightarrow {}^4He + 2p$.
- D-T fusion (${}^2H + {}^3H \rightarrow {}^4He + n + 17.6$ MeV) is the most promising power source. Requires ~10^8 K plasma confinement.
- Fusion stops at iron — beyond Fe, fusion costs energy. (Use to answer 'up to which element is fusion possible?' — iron.)

Closed-book recall prompts

- Up to which element is fusion energetically favourable? Why? (PS3 Q3 — exact.)
- Estimate the energy released per fission of ^{235}U.
- Write the chain reaction formula. Define critical / supercritical / subcritical.
- Why are light nuclei (water, graphite) used as moderators?
- Estimate the Coulomb barrier for D-T fusion. Compare to ~1 keV thermal energy.
- Sun lifetime: with N_0 hydrogens at luminosity L, find remaining time (subtract neutrino losses). (PST5 Q1 — exact.)
- Dose-based proton lifetime limit: 5 Gy/year, body ~70 kg, each decay deposits ~$m_p c^2$ → find τ. (2024/25 Q3(b) — exact.)

Common traps

- Forgetting to subtract neutrino energy losses when computing usable solar energy.
- Mixing up activation energy (the barrier height) with Q-value (the energy released).
- In dose calculations, forgetting unit conversions (MeV → J, per decay → per kg per second).

5. Past papers organised by year

If you'd rather see what each year tested as a whole, here it is. Use this when planning Phase 4 timed papers — pick a year, attempt under timed conditions, then mark.

2020/21 (PHAS0040A6UC)

- **Q1(a) [14 marks]** allowed/forbidden + Feynman (Λ^0, ψ^*, ve + p, μ^+e^- reactions)
- **Q1(b) [6 marks]** $D^0 \to K^-\pi^+$ decay-at-rest, find pion energy
- **Q2(a) [5 marks]** $H \to XX$ kinematics from natural width Γ, find X flight distance
- **Q2(b) [4 marks]** branching ratio $Br(\Phi \to u\bar{u})$ with colour factor
- **Q2(c) [4 marks]** gluon fusion x_1, x_2 at LHC for Y = 260 GeV
- **Q2(d) [2 marks]** fusion energy $^3H+^3H \to {}^4He+2p$ from BEs
- **Q2(e) [5 marks]** shell model spin/parity of ^{72}Ga from neighbours
- **Q3(a) [8 marks]** SEMF: does ^{75}Ge undergo β^+ or β^-? Energy released?
- **Q3(b) [10 marks]** HW production at LHC: count + ATLAS detection + b-jet kinematics
- **Q3(c) [2 marks]** minimum electron energy to resolve 10^{-18} m

2021/22 (PHAS0040A6UD)

- **Q1(a) [8 marks]** neutrino oscillations: estimate $\sin^2(2\theta)$, survival probability
- **Q1(b) [6 marks]** new heavy lepton: relative change in $\sigma(e^+e^- \to \text{anything})$ and R
- **Q1(c) [6 marks]** particle U decays via gluon and photon — deduce B, L, quark content
- **Q2(a) [14 marks]** allowed/forbidden + Feynman (ve + p, e^+ + ve, ϕ, p+p, e^-e^- reactions)
- **Q2(b) [6 marks]** invariant mass + 3-momentum from given 4-vectors
- **Q3(a) [9 marks]** heavy X = 250 GeV → ZZ/WW: discovery channels and mass measurement
- **Q3(b) [9 marks]** use SEMF to show lowest mass A=64 nucleus is $^{64}_{28}Co$
- **Q3(c) [2 marks]** how surface of nucleus changes when A increases ×8

2022/23 (PHAS0040A6UE)

- **Q1(a) [14 marks]** allowed/forbidden + Feynman (π^0, ϕ, p+p, Λ^0, e^+ ve reactions)
- **Q1(b) [6 marks]** particle $X \to \tau + \bar{\nu}_\tau$ decays at rest, find mass of X
- **Q2(a) [10 marks]** neutrino oscillations: $\sin^2(2\theta)$, Δm^2, survival probability
- **Q2(b) [3 marks]** ^{14}C decays per gram per minute
- **Q2(c) [2 marks]** BE of ^2H from $p+p \to {}^2H + e^+ + \nu_e + 0.42$ MeV
- **Q2(d) [5 marks]** elastic neutron-nucleus collision: derive $E_f/E_i = ((A-1)/(A+1))^2$
- **Q3(a) [6 marks]** Feynman diagrams for D^+/D^0 decays (Cabibbo-favoured/suppressed)
- **Q3(b) [2 marks]** explain why $D^0 \to K^+\pi^-$ rate << $D^0 \to K^-\pi^+$
- **Q3(c) [2 marks]** estimate the ratio of decay rates
- **Q3(d) [2 marks]** Higgs at LEP: dominant Feynman diagram + max $\sqrt{s}$
- **Q3(e) [2 marks]** gluon-gluon fusion Higgs production
- **Q3(f) [2 marks]** dominant Higgs decay channel
- **Q3(g) [4 marks]** $H \to ZZ \to 4\ell$ event count from σ_H, $\int L$, branching ratios

2023/24 (PHAS0040)

- **Q1(a) [6 marks]** Z boson momentum from electron-electron opening angle
- **Q1(b) [14 marks]** allowed/forbidden + Feynman ($n \to \pi^+\pi^-$, Ω^0_c, $e\mu$, $e\tau$, ϕ reactions)
- **Q2(a) [4 marks]** radiation length X_0 from electron stopping distance

- **Q2(b) [6 marks]** neutrino oscillation survival probability
- **Q2(c) [6 marks]** BE of ^{64}Zn from atomic mass + SEMF prediction, compare
- **Q2(d) [4 marks]** ^{14}C dating
- **Q3(a) [13 marks]** $e^+e^- \to \mu^+\mu^-$: total cross section + A_FB derivation
- **Q3(b) [7 marks]** W decay branching ratios via lepton universality + $W^+W^- \to e^+e^-\nu_e \bar{\nu}_e$

2024/25 (PHAS0040A6UG)

- **Q1(a) [14 marks]** allowed/forbidden + Feynman
- **Q1(b) [6 marks]** $B^0 \to D^-\pi^+$ decay-at-rest kinematics
- **Q2(a) [10 marks]** LHC parton kinematics + jet opening angle + jet measurement
- **Q2(b) [8 marks]** Feynman diagrams for B-meson decays + identify interaction + decay rate
- **Q2(c) [2 marks]** parity violation evidence (bookwork)
- **Q3(a) [11 marks]** bookwork: definitions (isotope, β^+, etc.) + shell model evidence
- **Q3(b) [9 marks]** proton lifetime from radiation dose limit

Good luck. The course is more compact than the write-up makes it look ☺